Ahmed Hassan Muhamad

Embedded Systems Engineer

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Summary

Embedded Systems Engineer with strong experience in embedded Linux, PCB design, analog circuits, and product development. Skilled in designing end-to-end embedded solutions from concept to production, with additional expertise in R&D and IoT systems.

Education

B.Sc. in Electronics and Communications Engineering -Thebes Academy 2019 – 2025

Technical skills

- Embedded Systems: Embedded C/C++, Embedded Linux, firmware development
- Hardware Design: PCB design, analog & digital circuit design, hardware debugging
- Communication Protocols: UART, I^2C, SPI
- Programming Languages: C, C++, Python, Bash
- AI & Automation: Embedded AI integration, sensor fusion, IoT systems, autonomous systems
- MCUs & Platforms: AVR, ESP32, Raspberry Pi, Arduino, VS Code, Proteus, Tinker Cad

Courses

- Embedded Systems Standard Diploma: C, Embedded C, AVR Microcontroller Interface & Communication Protocols
- ESP and Arduino Boards Programming and interfacing
- Embedded Linux Diploma: Python, C++, Linux Admin, Bash, Raspberry Pi
- PCB Design with KiCAD
- Electronic Circuit Design Using Altium Designer

Experience

National Energy Control Center (NECC) (Jul. 2023 -Aug. 2023)

Training

Future of Egypt Authority for Sustainable Development (Mar. 2025 -Mar. 2026)

- **Electrical Infrastructure Supervision (New Delta Project - 2,200,000 Acres):** Supervised full execution of MV pole erection, stringing, underground duct banks, and cable installation; verified phasing, terminations, and energization protocols.
- **MV Distribution Equipment:** Oversaw procurement, delivery, installation, and commissioning of Ring Main Units (RMU), MV switchgear panels, and associated protection relays in compliance with IEC standards.
- **Power Transformers (MV/LV):** Managed installation, oil treatment, insulation resistance testing, vector group verification, and staged energization of distribution transformers at multiple substations.
- **Low Voltage (LV) Systems:** Supervised complete LV works including cable routing/pulling, lugged terminations, MDB & Sub-MDB installation, LV transformer connection, cable tray/ladder systems, earthing, and final energization.
- **Consultant Coordination & Document Control:** Led technical submittal cycles with the consulting engineer; reviewed and approved shop drawings, material submittals, and method statements.

Freelancing Projects

- **DDDAS:** Graduation Project *(Grade: Excellent, showcased at Thebes Academy Projects Exhibition*): DDDAS (Deaf and Dumb Driving Assistant System) Developed an intelligent driving system enabling deaf and mute individuals to experience fully autonomous driving and integrate seamlessly into society. The project features multiple custom-designed AI models for real-time driver–passenger chat, road sign recognition, and auditory environment simulation to interpret surrounding sounds. A dedicated graphical user interface (GUI) was designed specifically for the system, which also supports satellite-based autonomous driving. Implemented using C, C++, Python, and Bash, with integration of UART and I^2C communication protocols.
  Read more: DDDAS
- **Smart Pills Box:** An embedded IoT medication management device developed on ESP32-C3. Designed custom PCB and 3D mechanical enclosure using KiCad, Proteus, and SolidWorks. The system schedules up to 10 medication slots with configurable alert windows, real-time RTC (DS3231) tracking, and simultaneous visual (WS2812 LEDs Units via FastLED) and audio alerts (DFPlayer Mini). Implemented precise battery voltage monitoring with charging detection, adaptive calibration, and low-pass filtering for stable readings. Built an integrated captive web portal (Access Point) for wireless configuration of medicine names, times, and alert durations using Arduino IDE. All settings are saved persistently via Preferences library. The device ensures reliable medication adherence through multi-modal reminders and user feedback via button and OLED display (SH1106).
  Read more: Smart Pills Box
- **Smart Timer Relay Board**: Electronic timer relay board designed to control high-voltage appliances (220–230 V AC) through commands received from a computer software via API
  Read more: Smart Timer Relay Board
- **Voltage Guard Pro**: This is a smart, low-cost power protection device designed to safeguard household appliances from dangerous voltage fluctuations. Built around the ATtiny85 microcontroller, it continuously monitors the AC mains voltage and instantly disconnects connected devices when the voltage goes below 190V or exceeds 250V protecting sensitive electronics from damage.
  Read more: Voltage Guard Pro
- **Smart Bottle-Top Dispenser**: The Bottle-Top Dispenser is a precision laboratory device designed to automate the dispensing of controlled liquid volumes. It provides accuracy, repeatability, and safety for handling chemicals in lab environments.
  Read more: Smart Bottle-Top Dispenser

language skills

- Arabic
- English